

CO1_AT2: ANALYTICAL

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Student Submissions — Analytical

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Student 39 **Vaibhava Lakshmi K** 192511310RD

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Q1. Harshad number

ANSWER

```
n=int(input("enter a number:"))
temp=n
sum=0
while temp>0:
    digit=temp%10
    sum=sum+digit
    temp=temp//10
if n%sum==0:
    print("harshad number")
else:
    print("not a harshad number")

print("harshad number from 1 to 500:")
for i in range(1,501):
    temp=i
    s=0
    while temp>0:
        digit = temp%10
        s=s+digit
        temp=temp//10
    if i%s==0:
        print(i,end=" ")
```

Q2. GCD

CODE

```
a=int(input("Enter first:""\n"))
b=int(input("Enter second:""\n"))
while b!=0:
    temp=a
    a=b
    b=temp%b
print("GCD is:",a)
```

INPUT

24
18

OUTPUT

Enter first:
Enter second:
GCD is: 6

Q3. sum to a target value

ANSWER

```
def find_pairs(lst, target):
    seen=set()
    pairs=[]
    for num in lst:
        complement=target-num
        if complement in seen:
            pairs.append((complement,num))
        seen.add(num)
    return pairs
data=[int(x) for x in input("Enter numbers:").split()]
t=int(input("Target:"))
result=find_pairs(data,t)
print("Pairs:",result)
```

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Q4. Bitwise Operators

ANSWER

```
a=0b1101=13(int)
b=0o25=21(int)
c=0x1A=26(int)
result=(a&b)|c
      =(13&21)|26
      =5|26
      =31(int)

output:
a=13,b=21,c=26,result=31
0b11111 0o37 0x1F
7 62 -32
```

Q5. binary search

CODE

```
def rotated_binary_search(arr, target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]<=arr[mid]:
            if arr[low]<=target<arr[mid]:
                high = mid-1
            else:
                low=mid+1
        else:
            if arr[mid]<target<=arr[high]:
                low=mid+1
            else:
                high=mid-1
    return -1
data = [4,5,6,7,0,1,2]
t=int(input("Target: "))
result=rotated_binary_search(data,t)
print("Found at:",result)
```

INPUT

6

OUTPUT

Target: Found at: 2